

Lecture 11

*Segmentation:
dividing the image in regions of interest.*

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Contents

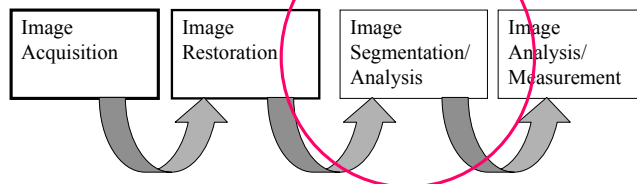
- Region based segmentation
 - Thresholding
 - Bi-level thresholding
 - Region growing
- Edge based Segmentation
 - Edge detection
 - Canny edges
- Optimization strategies

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Image Processing Workflow

- Now we look at sampled images
- Analyze images

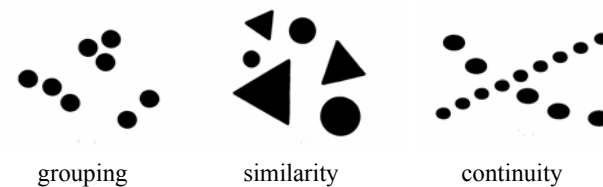


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Grouping

- Segmentation co-occurs with Grouping.
- Grouping is the process of associating similar image features together.
- Human vision & Gestalt:



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Segmentation, definitions

- **Segmentation** is the process of dividing an image in foreground and background
- **Segmentation** is to subdivide an image into its constituent regions and/or objects
- **Segmentation** is the process of dividing an image into regions of “related content”.
- **Segmentation** transforms an image of densities to a binary image
- Fact: there is no perfect segmentation technique. A technique is selected that fits the data!

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Segmentation, approach

- Segmentation algorithms are based either one of the basic properties of intensity values:
 - **Discontinuity**:
 - partitioning an image based on abrupt changes in the intensity: **edges**
 - **Similarity**:
 - partitioning and image into **regions** that are similar according to a set of predefined criteria.
- Region approach versus Edge approach

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Segmentation in Microscopy

- Images are flat scenes
- Ideal case: “even” illumination (flat field)
- Specific contrast has been applied to enhance the response to the detector
- 2D images
 - Range of techniques depending on material
- 3D images
 - Often same techniques extended to 3D
 - Require special treatment, relation with imager
 - Sampling is of non-isotropic

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Region Based Methods

- Thresholding, selects pixels of intensity range.
 - Result of Thresholding is a binary image!
- Region growing, selects similar pixels
 - Split Merge and Grouping
 - SLAR images
 - Scene projection images
 - Seed pixel region growing
 - Bio-medical, semi-interactive
 - Planar images

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Thresholding

- Select a value in dynamic range of the image: θ
- If $P(i, j) \geq \theta$ then $P(i, j) = \text{object} = 1$
Else $P(i, j) = \text{background} = 0$;
- Here θ , is the brightness threshold, i.e. light objects on a dark background. This is typical for Fluorescence Microscopy.
- Otherwise, for Bright Field Microscopy:
- If $P(i, j) < \theta$ then $P(i, j) = \text{object} = 1$
Else $P(i, j) = \text{background} = 0$;

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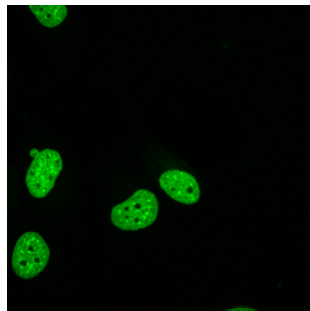
Threshold, selection

- Just try a value ...
- Examine the histogram
 - Uni-modal
 - Bi-modal
 - Multi-modal
- The isodata-method, heuristics
- The triangle-method
- The entropy method, heuristics
- Bi-level threshold
 - Density slicing

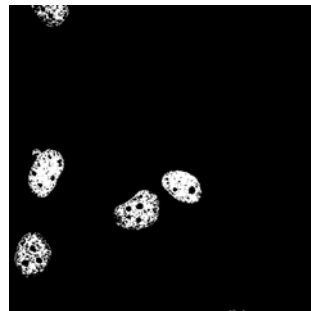
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Thresholding



Density, brightness image

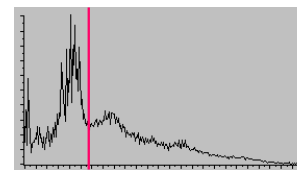


Binary image, some threshold

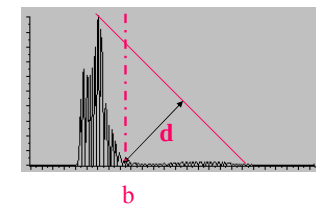
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Thresholding using histogram



Brightness value **b** is derived from %O under Histogram
Alt.: median, average, percentile

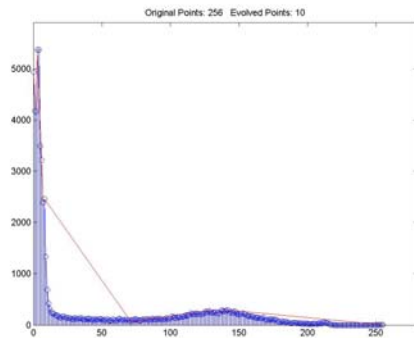


Triangle Method
Distance **d** is maximized to find Brightness value **b**.

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Bimodal Histogram



Typical histogram for Brightness Thresholding

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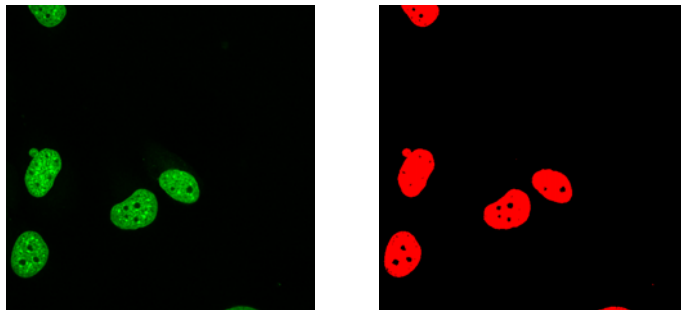
Thresholding, Isodata

- Developed by Ridler and Calvard
- Start with threshold value $T^0 = 2^{p-1}$,
– i.e. half dynamic range of image
- 1. Compute sample mean of foreground: m_f
- 2. Compute sample mean of background: m_b
- 3. $T^1 = (m_f + m_b)/2$...step 1
- 4. Stop : $T_i = (m_{f,k-1} + m_{b,k-1})/2$, $T_i = T_{i-1}$

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Result Isodata

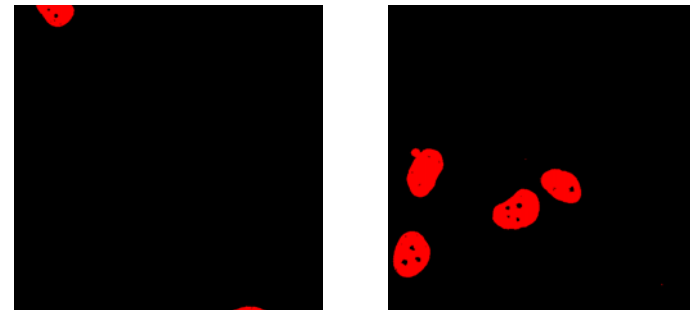


Found Isodata value = 73

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Additional processing required



Find object touching border

Xor with the result

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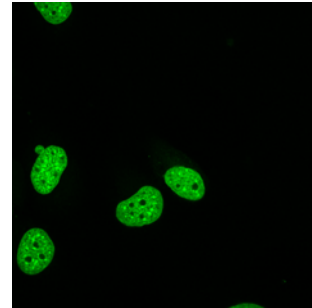
Thresholding, Entropy

- Entropy represent information richness, high number indicates that dynamic range is fully used.
- Compute the histogram
- Entropy of the histogram is used:
 - $f(i)$
 - f_i is the frequency of the i^{th} entry
 - $m \times n$ is the number of pixels
- Entropy is then:
 - $E = \sum_i -f(i) \cdot \log(f(i))$
 - This is minimized, taking fraction of the image pixels assigned to the foreground into account (of $m \times n$ imels).

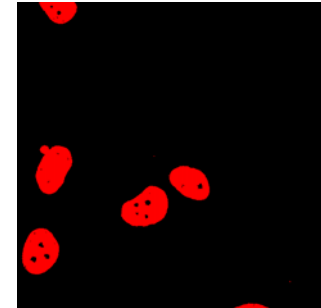
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Result Entropy



Entropy, from histogram



Using 0.05 as minimal fraction attributed to foreground. Entropy

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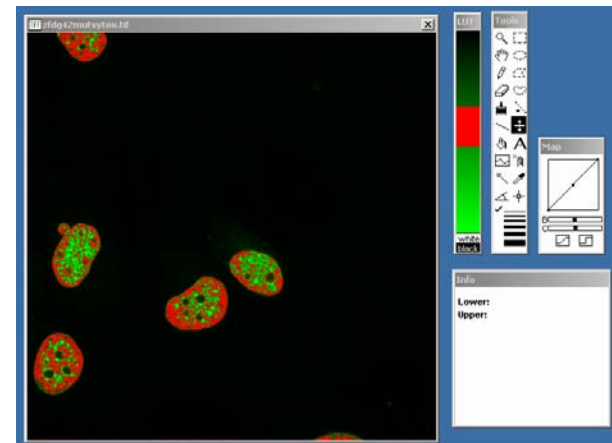
Bi-level Threshold, density slice

- Select lower threshold T^1
- Select upper threshold T^2
- If pixel $P(i,j)$ satisfies
 - $T^1 \leq P(i,j) \leq T^2$
 - Then $P(i,j) == 1$
 - Else $P(i,j) == 0$
- This is often realized interactive
 - Using the histogram
 - Using a LUT (see example): density slicing
 - Very popular in NIH image (Scion Image, ImageJ)

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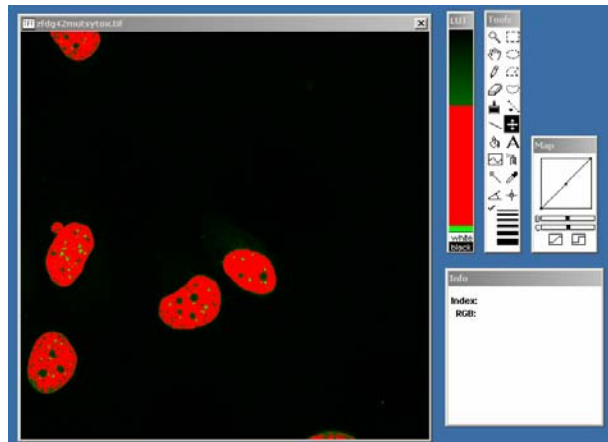
Result bi-level threshold (1)



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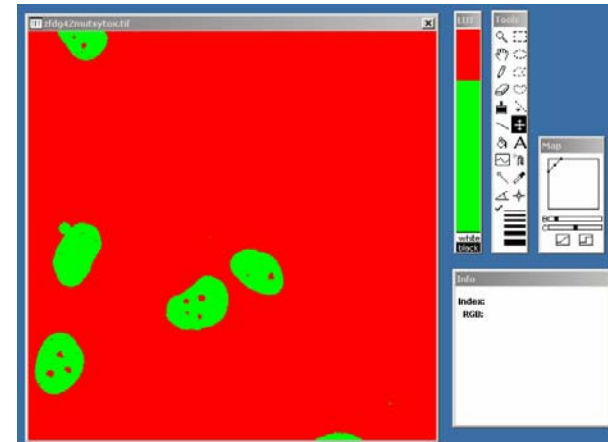
Result bi-level threshold (2)



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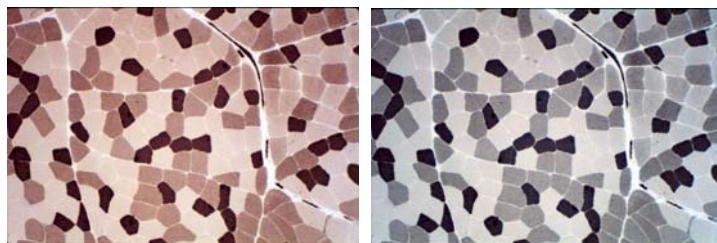
Result bi-level threshold (3)



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Thresholding in Bright-Field



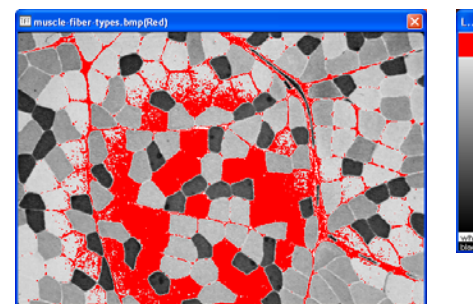
muscle fiber characterization
Image: specific contrast per fiber type

Grey-value representation
CCD capture.

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Segmentation, T-value selection (1)

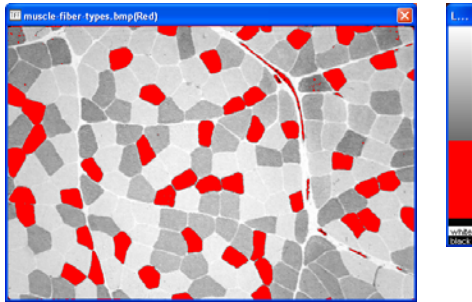


Select the bright muscle fibers?
Do we have all "unstained" fiber types?:
does the image need more preprocessing?

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Segmentation, T-value selection (2)



Select the dark muscle fibers?
Do we have separate fibers?: additional processing required

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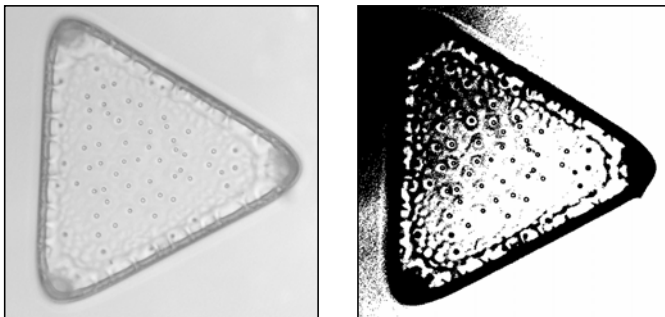
Thresholding, limitations

- The illumination of an image is not uniform and that affects the applicability of the threshold: **adaptive thresholding**.
- Uniform regions are selected. No relation between the regions is established.
- Sensitive to noise, or local disturbance. This prevents otherwise consistent regions to be expressed.

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Example uneven illumination

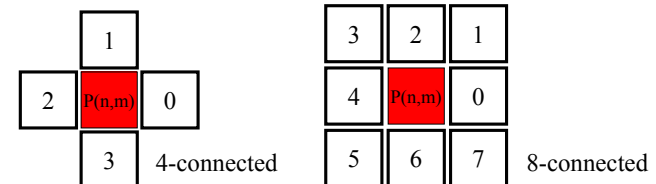


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Region growing

- A seed pixel represents:
 - Density value
 - Location
- What similarity criterion is chosen
- What connectivity is chosen



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Region growing, (simple) criteria

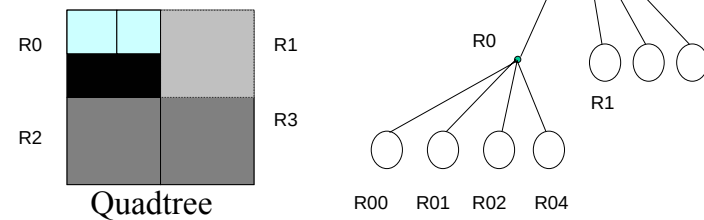
- If $T^{\text{low}} > \text{Neighbor} \geq T^{\text{high}}$
- Add Neighbor to Region R, grow
- Else proceed to next Neighbor
- Growth stops if Neighbors are outside given range.
- T^{low} and T^{high} are derived from T^0
- With T^0 , variance V is indicated for region R
- Variance V referred to as Tolerance for growing
- Region R should express Homogeneity
- Criterion = absolute difference with $f(x,y)$ of seed pixel

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Region growing, Split & Merge

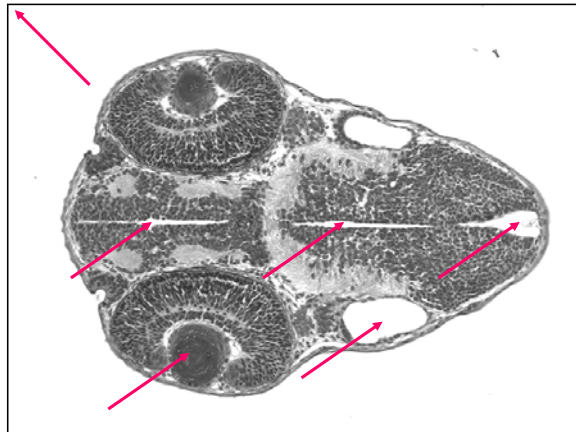
- P is homogeneity criterion/function
- Region R_i is inhomogeneous thus ($P(R_i) = \text{False}$) consequently is split into four sub regions
- Two adjacent regions R_i, R_j are homogeneous thus ($P(R_i \cup R_j) = \text{TRUE}$): they are merged



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E.G. Region growing with Seed Pixel



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Seed in background



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Seed in object

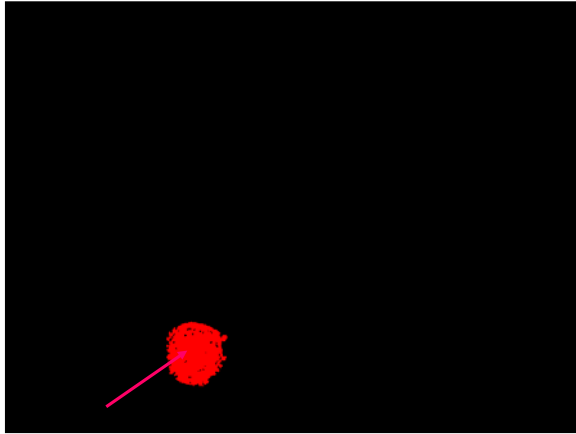


fig. Image Analysis in Microscopy 2006

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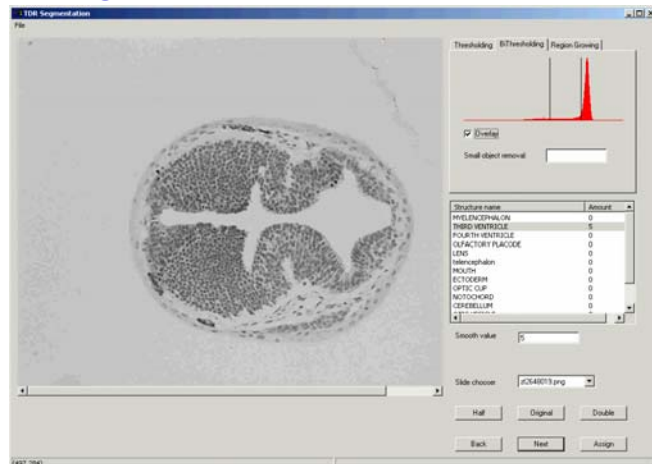
Result of Region Growing



fig. Image Analysis in Microscopy 2006

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Segmentation & Annotation



Example of semi-automated segmentation

fig. Image Analysis in Microscopy 2006

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Edge Based Methods

- Find edges in images using edge-operators
 - Sobel
 - Roberts
 - Prewitt
 - Marr/Hildreth
 - Canny
- These are susceptible to noise
- Suppress noise with Gaussian filter
- Suppressing noise is main feature of Canny edge detector

fig. Image Analysis in Microscopy 2006

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Edge, a definition

- An Edge is a set of connected pixels that are on the boundary between to regions.
- The edge is the location where the change occurs.
- Ideally, an edge defines a well behaved boundary between regions.
- Luminance of the scene is not ideal!
And detector introduces noise

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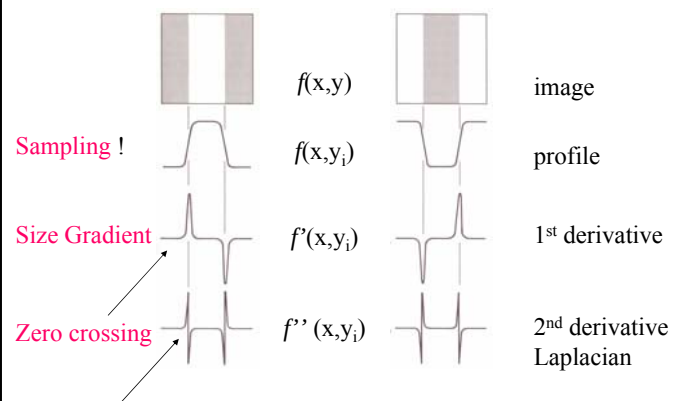
Edges, types

- Step edge 
- Ramp edge 
- Noisy edge 

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Edges and Operators



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Gradient

- The gradient is a vector
- Magnitude and Direction
- Magnitude: Edge strength
- Noise edge has high gradient: Smoothing !
- Gradient has component in x- and y- direction

$$\nabla I = (I_x, I_y) = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right) \quad \text{is the gradient}$$

$$\|\nabla I\| = \sqrt{I_x^2 + I_y^2} \quad \text{is the magnitude of the gradient}$$

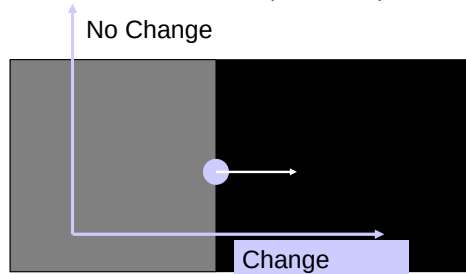
$$\alpha = \tan^{-1} \left(\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} \right) \quad \text{is the direction of the gradient}$$

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Visualizing the gradient

$$\left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right) = (k, 0)$$

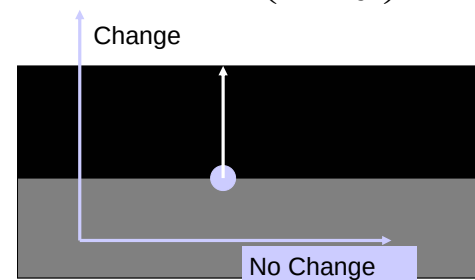


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Visualizing the gradient

$$\left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right) = (0, k)$$



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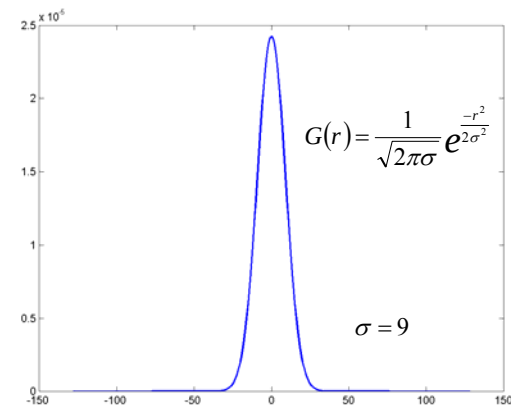
Canny Edges

- **John Canny** (1986) IEEE Transactions Pattern Recognition and Machine Analysis
- Filters are response to edge
 - step edge, noise edge, ramp edge
 - 1st derivative convolved with Gaussian identical to:
 - convolution with 1st derivative of Gaussian
- Two derivatives, in x and y direction (Gaussian)
 - good response to single edges
- Use a derivative of Gaussian filter (DoG)
 - Magnitude of Gradient, Direction of Gradient
- Use a Laplacian of a Gaussian filter (LoG)
 - Zero crossings indicate location of edge (maximum).

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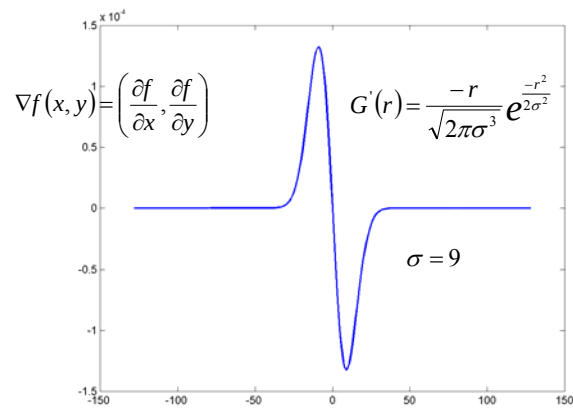
Gaussian: smoothing filter



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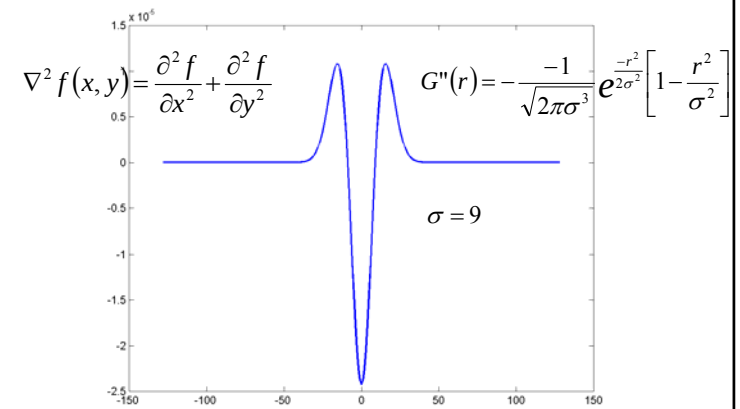
Derivative of Gaussian



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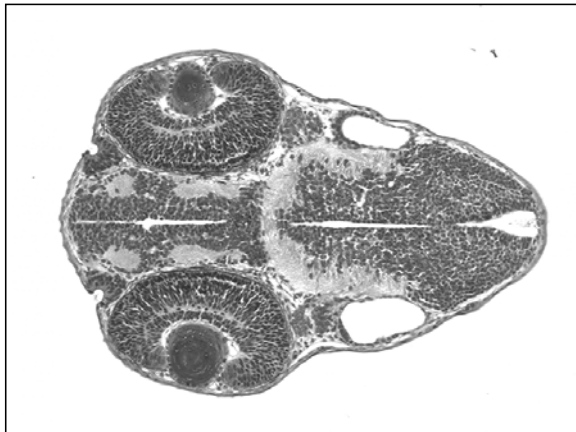
Laplacian of Gaussian



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E.G. with Gaussian, DoG and LoG



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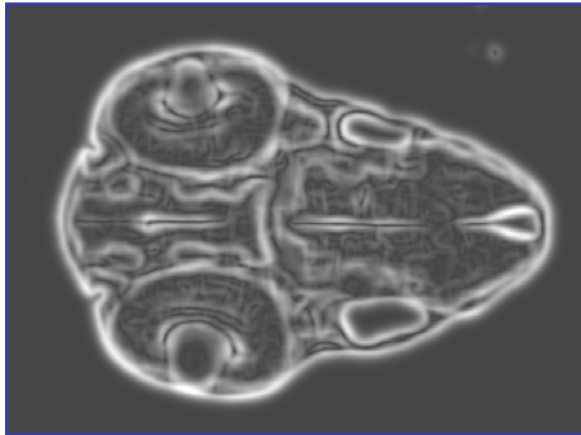
Recipe, Canny edges

- Compute gradient in X, using Gaussian
- Compute gradient in Y, using Gaussian
- Compute magnitude of gradient
- Compute Laplacian of Gaussian
- Establish Zero crossings
- Multiply zero crossings with Magnitude
- Threshold edges: non-maxima suppression
- Detail in result determined from
 - Width of Gaussian (σ) and
 - Threshold (edge strength)

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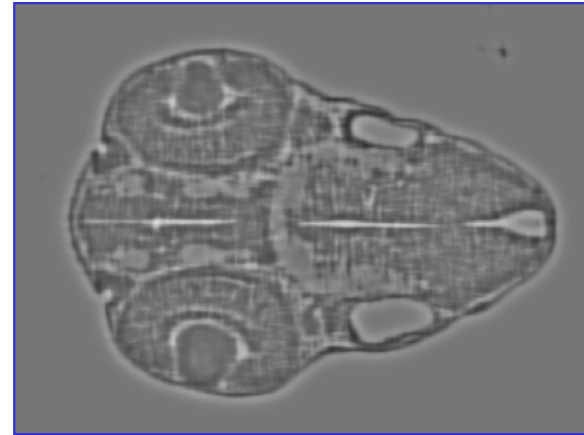
Compute Magnitude of Gradient



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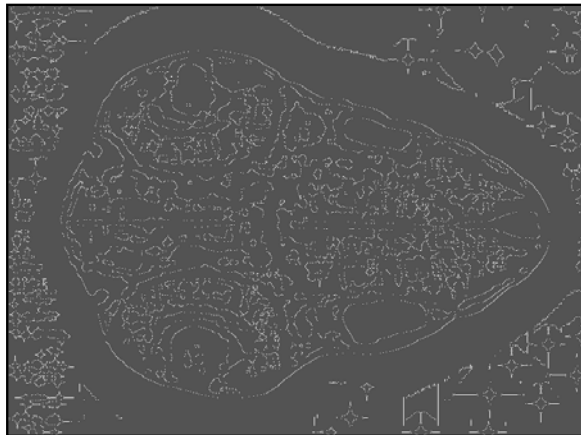
Compute 2nd derivative, LoG



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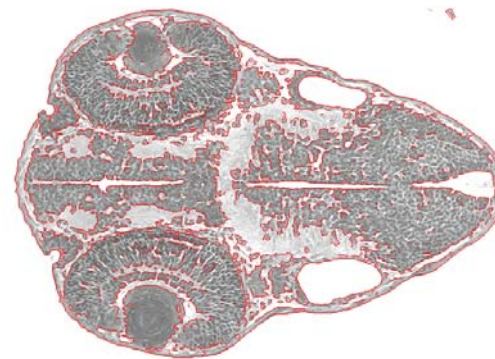
Zero Crossings, 2nd Derivative



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Canny edges: large sigma.



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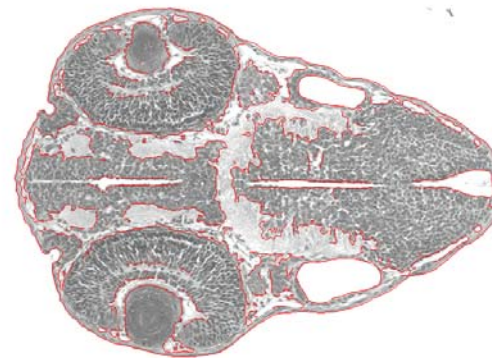
Canny edges: higher Threshold



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Canny edges: smaller sigma



A higher threshold filters stronger edges

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Optimization

- A model is optimized using form of heuristics an initial guess: a contour.
- Heuristics are used in relaxation of the contour to an “optimum”.
- Process of relaxation is iterative.
- During the process the connectivity of the initial guess (the model) is preserved.

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Optimization Methods

- Path searching with dynamic programming:
 - Contour: costs minimizing in matrix.
- Active contour model:
 - Snake: energy minimizing.
- Active shape model:
 - Model: evaluating the model
 - Set of predefined shapes & training images
 - Principal component analysis

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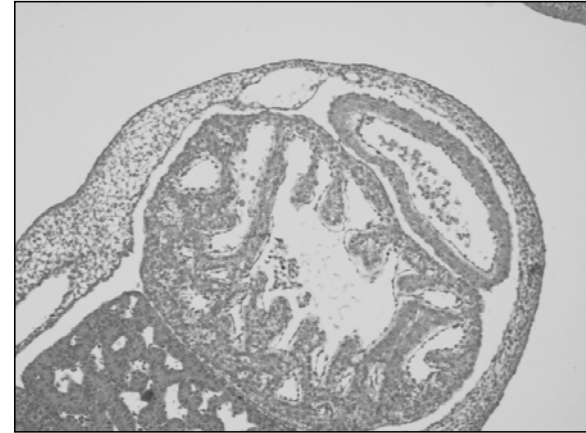
Minimal Cost Path

- Initial guess is a first estimate of the ideal contour describing a boundary.
- Perpendicular to the this initial guess the boundary is re-sampled with a width W .
- W together with the contour length, L , spans a cost matrix: $W \times L$.
- Find path which minimizes costs in $W \times L$
- Back projection of the path in image.

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Image with boundaries

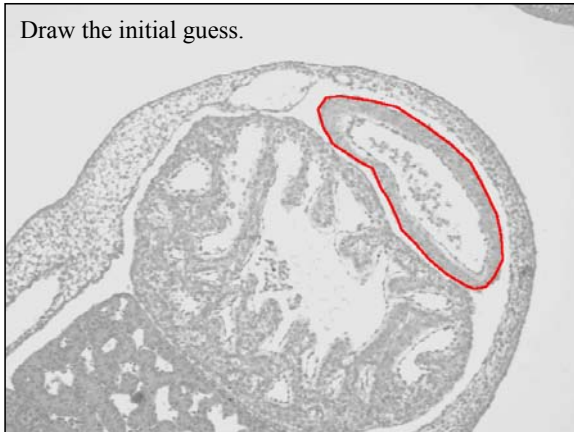


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Initial guess

Draw the initial guess.

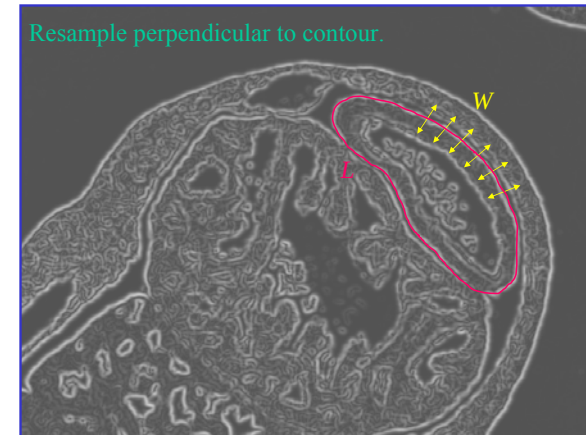


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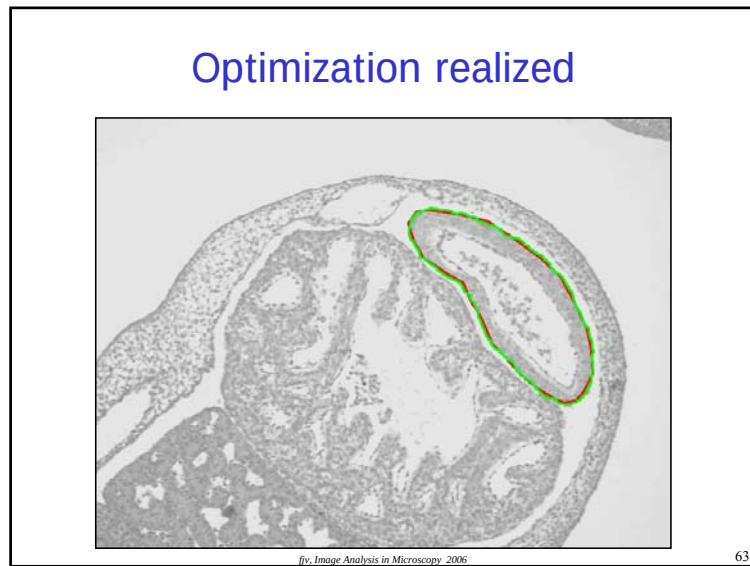
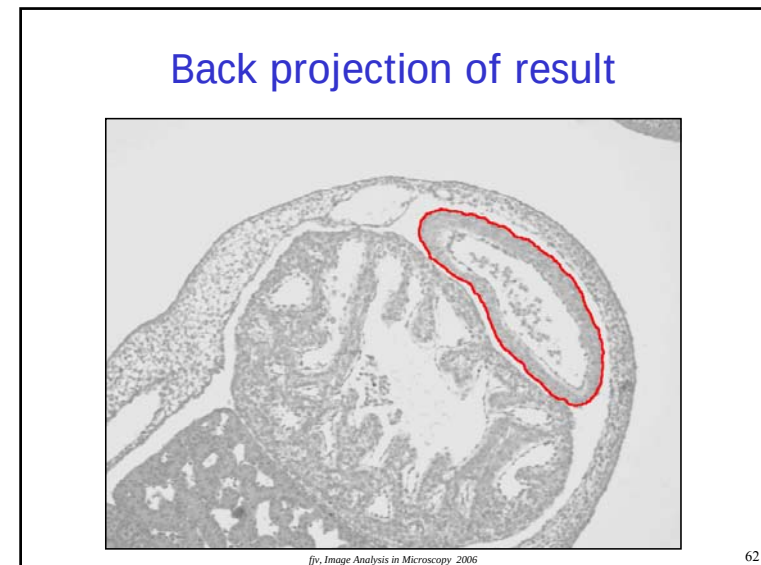
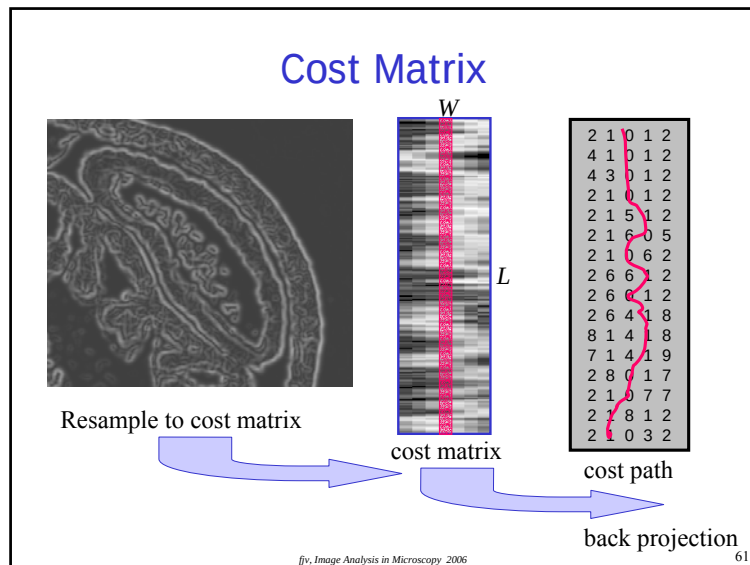
Computation of cost matrix

Resample perpendicular to contour.



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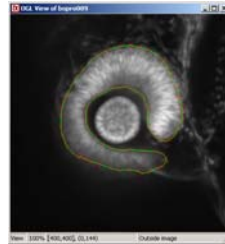
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- ### Application DP in Segmentation
- Successful with
 - Noisy images
 - Propagation of result
 - Rough estimate of a delineation is available
 - Figure of Merit, Cost Function
 - first derivative, magnitude of the gradient
 - distance function
 - bonus (different functions)
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Active Shape Model (ASM)

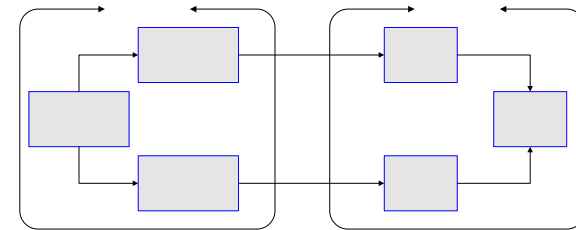
- Anatomical structure is the “shape”
- Shape consists of n points, i.e. Landmarks: point set
- Shape model is based on a Principal Component Analysis (PCA) for modelling



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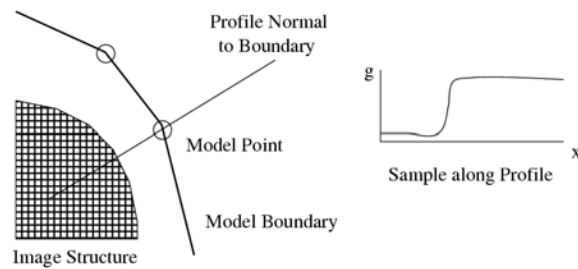
Building an Active Shape Model



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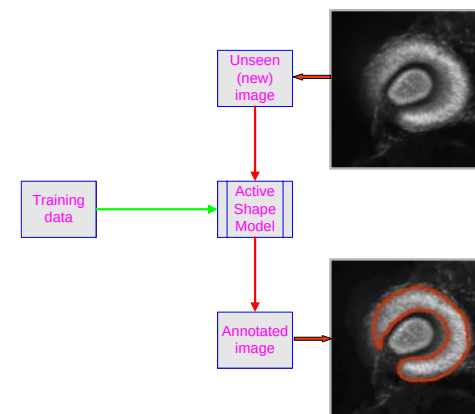
Landmark Points



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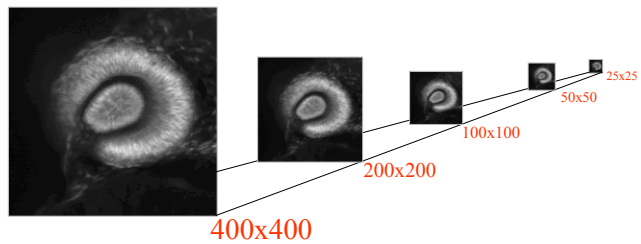
Application Active Shape Model



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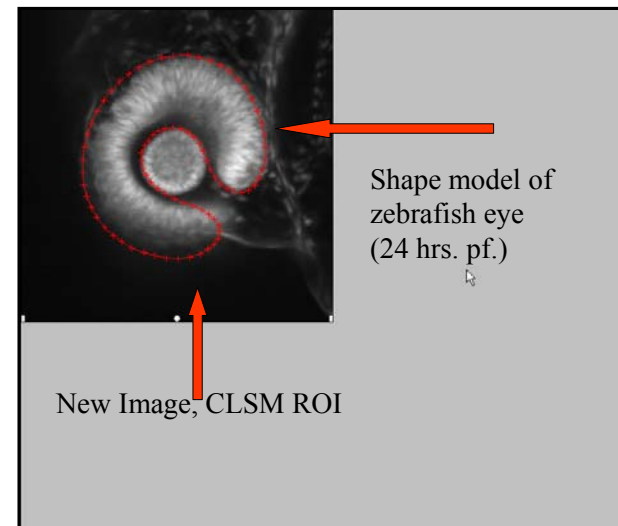
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Resolution pyramid



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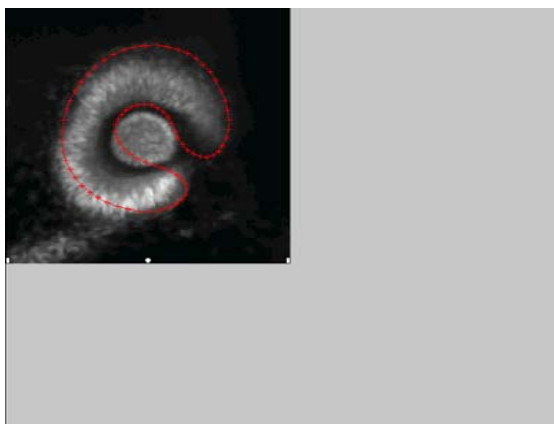
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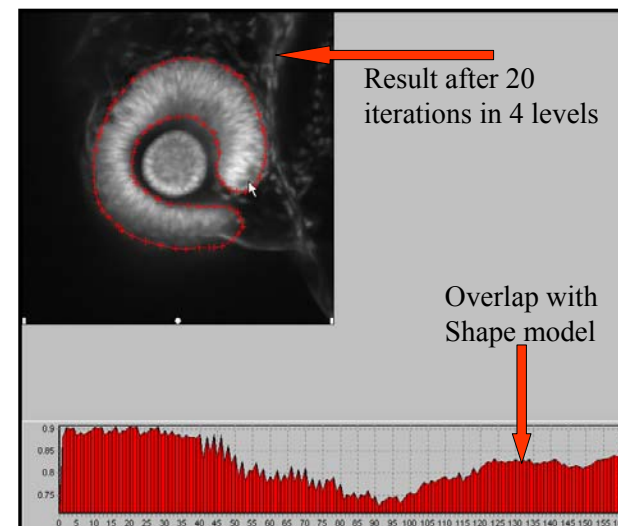
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Example of ASM application



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Review Lecture 11



- Segmentation
- Thresholding
- Region growing
- Edge detection
 - Magnitude of the Gradient
 - Direction of the Gradient
 - Concept of Canny edge detector.
 - Gaussian, DoG and LoG
- Optimization with Minimal Costs, ASM

Segmentation revisited

- Binary image processing
- Motion analysis
 - Optical flow
- Distance transformations (lecture 12)
 - Watershed segmentation
- Hough transformations
 - Line detection
- Texture analysis
 - Texture segmentation

Acknowledgements

Books

- Gonzales & Woods: Digital Image Processing

Papers

- Canny, 1986

Other

- FJV, test and experimental data
- Jacobs (NEC), gradient examples